

Akash Kumar

PhD Scholar, Geospatial Engineering
Indian Institute of Technology Roorkee

🏛 Civil Engineering Dept., IIT Roorkee
📍 Roorkee, India - 247667
✉ akash_k@ce.iitr.ac.in
🌐 akashpandey.com
📞 +91-9045865902

Education

2022 – 2026

Ph.D.

Geospatial Engineering – Indian Institute of Technology Roorkee

Thesis: Multi-scale analysis of crop phenology using PhenoCam & satellite data

| Submitted initial dissertation

2020 – 2022

M.Tech

Geoinformatics – Maulana Azad National Institute of Technology, Bhopal

Thesis: A web-GIS based approach for mapping rice residue burning and generated pollutants over Punjab

| Branch Gold Medalist; 9.51 CGPA

2014 – 2018

B.Tech

Civil Engineering – Meerut Institute of Engineering & Technology, Meerut

Experience

2023 – Present

Bhoomicam Pvt Ltd 📍 IIT Roorkee

Co-Director

Handling operations and B2B partnerships.

Apr 2025 – Present

Chief Remote Sensing Scientist

Led the technical team in building geospatial solutions for crop monitoring.

Jun 2023 – Apr 2025

2018 – 2019

Civil Site Engineer – BHEL 📍 Yelahanka, Bengaluru

Supervised construction activities, quality control, and coordinated designs with on-site teams.

Publications

Peer-Reviewed Journals

- [1] **Akash Kumar** and Siddhartha Khare (2026). Evaluation of PlanetScope and Sentinel-2 Derived Wheat Crop Phenology Using Infrared-Enabled PhenoCam and Field Observations. *Ecological Informatics*, 103821. [doi:10.1016/j.ecoinf.2026.103821](https://doi.org/10.1016/j.ecoinf.2026.103821).
- [2] **Akash Kumar**, Siddhartha Khare, and Sergio Rossi (2025). PhenoAI: A deep learning Python framework to process close-range time-lapse PhenoCam data. *Ecological Informatics*, (88), 103134. [doi:10.1016/j.ecoinf.2025.103134](https://doi.org/10.1016/j.ecoinf.2025.103134).
- [3] **Akash Kumar**, Ganesh Sai Sivani Noolu, Siddhartha Khare, and Sergio Rossi (2025). A comparison of PhenoCam and satellite indices with in-situ observations for black spruce in the boreal forest of Quebec, Canada. *Ecological Informatics*, (88), 103137. [doi:10.1016/j.ecoinf.2025.103137](https://doi.org/10.1016/j.ecoinf.2025.103137).

International Conference

- [1] **Akash Kumar** and Siddhartha Khare (2026). An Automated Workflow for Extracting Land Surface Phenology from Phenocam Time-lapse Images. *Asia Oceania Geosciences Society (AOGS 2026)*. Accepted and presentation scheduled in August 2026. Fukuoka, Japan, 2026.
- [2] **Akash Kumar** and Siddhartha Khare (2026). Establishing PhenoCam-based Monitoring of Wheat and Rice Phenology in India with Satellite and Meteorological Data. *EGU General Assembly 2026*. Vienna, Austria, 2026. [doi:10.5194/egusphere-egu26-14674](https://doi.org/10.5194/egusphere-egu26-14674).
- [3] **Akash Kumar**, Noolu G. S. Sivani, Siddhartha Khare, and Sergio Rossi (2024). Linking Phenocam Derived Phenology With Satellite and Field Observations in the Evergreen Forest. *2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*. 2024. [doi:10.1109/InGARSS61818.2024.10984047](https://doi.org/10.1109/InGARSS61818.2024.10984047).

Book Chapter

- [1] Tushar Bharadwaj, **Akash Kumar**, and Siddhartha Khare (2023). Smart agriculture platforms: Case studies. *Technological Prospects and Social Applications of Society 5.0*. CRC Press, 2023, 169–176. [🔗 doi:10.1201/9781003324720-15](https://doi.org/10.1201/9781003324720-15).

Manuscripts Under Review & Preparation

- [1] **Akash Kumar**, S. Goswami, and Siddhartha Khare (2026). *Deep Learning-Based Field Segmentation and Multitemporal Sentinel-2 Analysis for Crop Yield Estimation in Smallholder Systems of India*. In preparation. 2026.
- [2] **Akash Kumar**, Zainab Khan, Siddhartha Khare, Umesh Kumar Saxena, and Purushottam Kumar (2026). Integrating PhenoCam and AWS Data to Quantify Stage-Specific Climatic Impacts on Rice Phenology. *Remote Sensing Applications: Society and Environment*. [🔗 doi:10.2139/ssrn.6087391](https://doi.org/10.2139/ssrn.6087391). Under review.

Conferences & Workshops

Attended & Presented

2026	 Attend	EGU General Assembly 2026 (EGU26) 📍 Vienna, Austria 🔗 post
2025	 Present	GeoSmart India Conference & Expo 📍 Bharat Mandapam, New Delhi 🔗 post
2025	 Attend	Workshop on Synthetic Aperture Radar (SAR) 📍 IIT Tirupati
2024	 Paper	India Geoscience and Remote Sensing Symposium (InGARSS) 📍 NIT Goa 🔗 post
2024	 Poster	International Conference on Geomatics in Civil Engineering 📍 IIT Roorkee
2023	 Attend	2nd Roorkee Water Conclave 📍 NIH Roorkee
2022	 Attend	Natural-Hazard Symposium for the Indian Himalayas 📍 IIT Roorkee

Delivered & Conducted

2026	 Lecture	<i>Working with Google Earth Engine</i> 📍 GNEC, IIT Roorkee Greater Noida Campus Invited hands-on lecture for IN-SPACe Short-Term Skill Development Course 🔗 post
2023	 Workshop	Modern Surveying Instrument Karyashala 📍 IIT Roorkee Teaching assistant
2022	 Training	Google Earth Engine 📍 NIGMT New Delhi Online
2021	 Training	Google Earth Engine 📍 MANIT Bhopal Water Resource Dept.

Fellowships & Grants

2026	 Funding	AOGS reduced registration fee support for the 23rd Annual Meeting Support to attend and present at AOGS 2026 📍 Fukuoka, Japan
2026	 Funding	IIT Roorkee Department Development Fund (DDF) ₹1,50,000 for conference presentation at EGU General Assembly 2026 📍 Vienna, Austria
2026	 Honorarium	NPTEL TA for 8-Week Course Repeat offering; ₹40,000 honorarium
2025	 Honorarium	NPTEL TA for 8-Week Course ₹1,00,000 🔗 TA list
2024	 Funding	InGARSS Travel Grant for Conference \$250 for IEEE InGARSS 2024 📍 NIT Goa
2022	 Award	First Prize Winner, National Geospatial Startup Competition by AABSyS ₹50,000 🔗 details
2022	 Scholarship	SRF & MHRD Scholarship for PhD
2020	 Scholarship	MHRD Scholarship for Post-Graduation

Teaching & Mentoring

Academic Courses

2023, 2024, 2025	 Spring	Geospatial Engineering-I (CEN-102) Dept. of Civil Engineering, IIT Roorkee
2022, 2023, 2024, 2025	 Autumn	Geospatial Engineering-II (CEN-201) Dept. of Civil Engineering, IIT Roorkee
2022	 Spring	Basics of Remote Sensing Technology (GI-103) MANIT Bhopal
2021	 Autumn	Digital Processing of Remotely Sensed Data (GI-202) MANIT Bhopal; online tutorials

Mentoring & Field Training

2025, 2026	 NPTEL	Digital Land Surveying & Mapping , NPTEL, IIT Roorkee Jan–Mar offerings; course coordination, uploads, assignments, and query resolution
2023–24	 Mentor	External Academic Mentor of Mr. Dipanshu Priyadarshi , Gati Shakti Vishwavidyalaya, Vadodara Spring term; UG project on Land Use Land Cover Change Detection of Vadodara District
2022 – 26	 Training	Multiple hands-on practical sessions for railways, land planning, and officer training programs at GNEC, Noida IIT Roorkee Extension Centre
Dec 2022	 Field Camp	Survey Camp for 2nd Year UG Students Led on-field surveying and managed 120+ students

Technical Skills

Software

Programming	Python, R
Data & Statistics	Advanced MS Office, SAS JMP Pro, LaTeX, MS Visio, SigmaPlot
Geospatial	Google Earth Engine, QGIS, ArcGIS Pro, ERDAS IMAGINE, iGIS
Photogrammetry	Pix4D Mapper, FARO Scene, CloudCompare
GPS & Geotagging	Emlid Studio, Spectrum Survey Office, QBase
Spectral Analysis	SVC's Spectra Vista, ENVI
Civil Engineering	Autodesk AutoCAD 3D, Revit Architecture, STAAD.Pro

Instrument Handling

UAVs	IdeaForge Q6 (RGB and multispectral), Trinity F90+ Hybrid VTOL (LiDAR and oblique RGB), DJI Matrice 100, Agri-Spray Drone (quad- and hexacopter)
Sensors	PhenoCam Camera, Automated Weather Station (AWS), Spectroradiometer SVC XHR-1024i
GPS	Trimble DA2, Emlid Reach RS2/RS3 RTK, Sokkia GRX2
TLS	FARO Terrestrial Laser Scanner
Surveying	Total Station (robotic and EDM), Analytical Theodolite, Digital/Auto/IOP/Dumpy Level

Professional Service

Peer Review

Ecological Informatics

25+ reviews

International Journal of Applied Earth Observation and Geoinformation

20+ reviews

Information Processing in Agriculture

5 reviews

Smart Agricultural Technology

2 reviews

Memberships

IEEE GRSS

Geoscience and Remote Sensing Society

IIM

Indian Institute of Metals

EGU

European Geosciences Union

AOGS

Asia Oceania Geosciences Society

Extracurricular Experience & Activities

2022 – Present	Represented Bhoomicam Pvt Ltd at startup expos, innovation events, and outreach programs. - NTRO AI Grand Challenge Hackathon 2025 New Delhi Shortlisted Stage 1; ₹ 8,00,000 details - Climate Innovation Summit 2025; Operation Dronagiri 2025; Defense Surya Drone Tech 2025 - NASSCOM Technology Confluence 2024 post; TIPS 4.0 2024; IISF 2023; KVK outreach around Roorkee
2025, Jul	Geospatial Mapathon 2025, IIT Bombay FOSSEE Winner results
2025, Jul	NRSC Offline Coding Challenge 2025, ISRO Second prize post
2024	UAV Pilot Certificate – IdeaForge Certified NP2317
2024, Oct	NASA Space Apps Challenge 2024 Participant post
2022 – 2024	Inter-Bhawan Chess Tournament, IIT Roorkee 3 gold medals 2022, 2023, 2024
2016, Mar	Guinness World Record Holder – World's Biggest Orchestra Record holder guitarist; Art of Living Foundation, New Delhi details
2014, Oct	UP Open Chess Tournament (U-17) Ranked #3

References

Prof. Siddhartha Khare

IIT Roorkee

siddhartha.khare@ce.iitr.ac.in

Prof. S. K. Katiyar

MANIT Bhopal

skatiyar7@rediffmail.com

Prof. Kamal Singh

MANIT Bhopal

singhk@manit.ac.in